

Understanding Pilots' Perceptions of AI-Mediated Mental Health Support in Aviation: A Socio-Technical Framework

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Abstract

Commercial pilots operate in safety-critical, highly regulated environments where mental health disclosure is closely tied to licensure and career risk. While prior work documents elevated psychological distress among pilots, little research examines how pilots perceive AI-mediated mental health support within this professional context. We present findings from semi-structured interviews with commercial pilots, analyzed using iterative thematic analysis. Our analysis introduces a layered socio-technical framework illustrating how regulatory constraints, organizational systems, operational demands, and personal identity interact to shape engagement with AI-mediated mental health support. Pilots described conditional acceptance of AI, positioning it as most viable when functioning as a trust-bounded, private outlet within the individual domain, while remaining cautious of tools integrated into organizational or regulatory systems. These findings demonstrate that adoption is shaped not only by stigma, but by concerns about institutional visibility and professional consequences.

Keywords

Human-Centered AI, Aviation Psychology, Pilot Mental Health, AI-Mediated Mental Health, Socio-Technical Framework, Trust in AI

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Human-centered computing ~ Human computer interaction (HCI)
~ HCI design and evaluation methods ~ User studies

Human-centered computing ~ Human computer interaction
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Computing methodologies ~ Artificial intelligence

1 INTRODUCTION

Commercial pilots operate in high-stress, safety-critical environments that demand sustained cognitive performance, emotional regulation [1], and continuous fitness to fly. Prior research documents elevated rates of depression, anxiety, fatigue, and psychological distress [3, 10] among pilots compared to the general population. These outcomes are closely linked to occupational stressors such as irregular schedules, circadian disruption, cumulative fatigue [9], and prolonged exposure to high-responsibility decision making, positioning pilot mental health as both a wellbeing and aviation safety concern.

Despite this burden, mental health challenges among pilots remain widely underreported [4, 7]. Studies show that stigma, fear of career consequences, and distrust in the confidentiality of aeromedical reporting [4] systems discourage pilots from seeking formal mental health care. In regulatory contexts, such as the United States, disclosure of psychological symptoms may trigger grounding [4] (suspension of a pilot's authorization to fly due to medical or regulatory concerns) or extended medical review, leading many pilots to delay care, avoid diagnoses, or rely on informal coping strategies instead [1]. These patterns suggest that barriers to mental health support are attitudinal and structurally embedded within regulatory and institutional constraints, where disclosure becomes professionally visible and carries career consequences.

In parallel, AI-mediated mental health tools [8] have expanded rapidly. Conversational agents and related systems have demonstrated potential as adjunct or preventative tools for managing mild [5] or subclinical mental health symptoms [6] in general populations, offering accessible and stigma-reducing support. Recent aviation-focused reviews suggest that such tools may align with pilots' preferences for discretion and self-reliance, particularly when

designed with strong privacy and governance safeguards [8, 11]. However, this work remains largely speculative with respect to aviation. To date, there is little empirical research examining how pilots themselves perceive AI-mediated mental health tools, or how such technologies are evaluated within the context of safety culture, institutional oversight, and documentation risk. As a result, efforts to introduce AI-mediated mental health interventions may overlook the professional and regulatory contexts that shape pilots' lived realities. This study addresses this gap by examining pilots' perceptions of mental health, existing support systems, and the potential role of generative AI-mediated mental health support in aviation contexts.

1.1 Research Questions

Guided by gaps identified in prior literature, we ask:

- What beliefs, concerns, and barriers shape how pilots' approach and engage with mental health support?
- How do pilots perceive the role of conversational AI in supporting mental health, and what factors shape their trust, openness, and willingness to engage with such tools?
- In which conditions and constraints might pilots choose to engage with or avoid generative AI designed to support emotional and cognitive wellbeing?

2 METHODOLOGY

2.1 Participants

This study draws on interview data from 31 pilots across two sites: University of Washington (UW; $n = 13$) and University of Melbourne (UMelb; $n = 18$). UW interviews formed the primary basis of analysis, as they explicitly examined pilots' perceptions of AI-mediated mental health support. Although thematic analysis was conducted on UMelb interviews, these data were collected for a different research purpose and did not focus on mental health or AI-supported wellbeing. Accordingly, UMelb data informed broader contextual understanding, while findings presented here are primarily grounded in the UW dataset.

UW participants were recruited through opportunity sampling via the research lead's professional networks. Interested pilots completed an online screener assessing eligibility, availability, and background characteristics (e.g., region, total flight hours, route type, and prior AI experience). Eligibility criteria included holding or having previously held an Airline Transport Pilot (ATP) certificate, English proficiency, and willingness to participate in a 60-minute recorded virtual interview. Of 56 pilots who completed the screener, 13 were selected. Participants ranged in age from 25 to 64 and were affiliated with commercial airlines across seven countries in Asia, Europe, and North America. Eleven participants identified as male and two as female. Twelve held an active ATP certificate, while one had lost medical clearance. Ten reported prior experience with AI tools, and three reported none.

2.2 Procedure

Semi-structured interviews were conducted over a three-week period using a protocol designed to examine pilots' lived experiences,

attitudes, and reasoning around mental health, disclosure, and AI-enabled support in aviation. Interviews were conducted via Zoom, lasted approximately 60 minutes, and were audio-recorded and transcribed for analysis. Participants were presented with two pilot-specific hypothetical scenarios (one low-intensity and one high-intensity) and asked to reflect on existing support resources and how AI-mediated mental health tools might assist in each context. One researcher led each interview, with one to two additional researchers taking notes.

2.3 Data Analysis

Interview transcripts were analyzed using an iterative thematic analysis approach using grounded theory [2]. Initial open coding identified salient experiences, concerns, and interpretive frames related to mental health, professional identity, trust, and technology use. Through iterative comparison across participants, codes were refined and consolidated into a structured codebook capturing recurring patterns as well as points of divergence.

The final analysis focused on relational synthesis, examining how professional risk, information traceability, coping practices, and perceptions of AI interacted across accounts. This process resulted in a socio-technical framework describing how pilots' reason about AI-mediated mental health support and the conditions under which such tools are considered acceptable or inappropriate within aviation contexts.

3 RESULTS

Across interviews, pilots evaluated mental health, coping practices, institutional resources, and generative AI technologies through shared professional contexts centered on regulatory exposure, information traceability, and accountability. Prior research has established elevated mental health risk among pilots [1, 6, 7] and persistent barriers to help-seeking associated with certification and career concerns [3]. Building on this literature, our analysis examines how these structural conditions shape pilots' interpretations of mental health engagement and emerging AI tools. We synthesize these findings into a socio-technical framework that captures how mental health and AI engagement are negotiated in aviation contexts.

3.1 Mental Health as Regulated Professional Risk

Across interviews, pilots consistently framed mental health in relation to licensure, employability, and professional identity. Medical certification was described as inseparable from livelihood and long-term career stability. As one participant explained, "Everything is related to one license. Your medical - it's part of your license. Your license is your way of living, your status, your future, your everything."

Within this framing, mental health disclosure was widely understood as carrying significant professional consequences. Many participants (10/13) described concerns that disclosure could lead to grounding, extended medical review, or loss of certification. One participant summarized this risk succinctly: "Yeah, your careers on the line... you say something wrong, or the wrong notes get put into your file. That's career-ending." Mental health was therefore

evaluated as a regulated exposure embedded within documentation and oversight systems rather than as a discrete personal concern.

Participants' accounts reflected variation across regulatory contexts. U.S.-based pilots emphasized the permanence of medical records and uncertainty surrounding future reinterpretation of disclosed information. European pilots described a greater normalization of psychological care within their regulatory and cultural environments. While mental health policies and reporting requirements vary across European countries, an ethos of openness toward mental health support was exemplified as one European pilot noted, "In the U.S., pilots are scared to get mental health support. And for me, as a European, it's normal to go to a psychologist."

These conditions shaped pilots' engagement strategies. Several participants described deliberate efforts to limit documentation or formal diagnosis as a means of managing professional risk. One participant articulated explicit boundaries around care: "You cannot diagnose me. You cannot give me any sort of diagnosis at all. I will pay cash out of my pocket... because you cannot submit to insurance without a diagnosis." These practices indicate that avoidance of formal disclosure functioned as a calculated response to regulatory uncertainty rather than disengagement from mental health concerns.

3.2 Institutional Visibility, Information Control, and Adaptive Coping

Concerns about licensure risk were closely tied to how pilots evaluated information traceability and institutional visibility. Engagement with mental health resources was shaped by who could access shared information, how it would be recorded, and whether it could later be used in certification or employment decisions. Participants described carefully managing documentation pathways. One pilot articulated strict boundaries around formal diagnosis: "You cannot diagnose me. You cannot give me any sort of diagnosis at all. I will pay cash out of my pocket... because you cannot submit to insurance without a diagnosis." Employer-provided mental health tools were frequently approached with skepticism, with many participants (10/13) expressing concern that information shared through company-linked systems could become accessible to management or regulators. Union-sponsored programs such as Professional Standards (ProStans) occupied a more trusted but still conditional space. As one participant explained, "It's pilot-to-pilot, and it stays out of the company system. It's a way to talk about issues without it immediately becoming a company or FAA thing." Another highlighted documentation differences: "If you go through ProStans, it's not documented the same way. It gives you a chance to deal with something before it becomes official." Even so, participants acknowledged that unions remain embedded within broader regulatory structures, limiting complete insulation. Together, these accounts illustrate how trust in mental health support was closely linked to control over information flow and the timing of formal documentation.

Within these regulatory and informational constraints, pilots described relying primarily on informal and self-directed coping strategies that avoided formal documentation. Exercise, structured routines, and spiritual or religious practices were commonly cited as ways of managing stress without generating professional exposure.

At the same time, alcohol and drug use was described by several participants as culturally normalized within aviation. One participant referenced "a very famous pilot support group called The Pub," noting that informal social spaces historically functioned as outlets for stress. These strategies were situated within a broader professional culture emphasizing endurance and continued operational performance. Most participants (10/13) referenced expectations to continue flying despite psychological strain, and some (3/13) delayed formal support to preserve certification status. As one pilot summarized, "I wanted to keep flying, so I just tried to deal with it." Rather than reflecting indifference toward mental health, these accounts illustrate how coping practices were shaped by efforts to manage institutional visibility and preserve employability.

3.3 AI as Trust-Bounded Reflective Support

Pilots evaluated AI-mediated mental health tools through the same lens of institutional visibility and information control described above. Nearly all participants (12/13) reported familiarity with contemporary AI systems such as ChatGPT, Copilot, Claude, or Gemini. These tools were primarily used for low-risk, task-oriented purposes such as drafting emails, generating checklists, or organizing information. As one participant noted, "I use it for emails almost all the time... It's rare I don't." In these contexts, AI was treated as assistive infrastructure, with decision-making authority remaining clearly with the pilot.

Beyond task-based use, a majority of participants (8/13) described AI as useful for reflection and sensemaking. Several characterized AI as a starting point for organizing thoughts rather than a source of authoritative guidance. One participant described it as "a place to start from," explaining that even when responses were imperfect, they provided structure for thinking through a problem. Others described value in having personal data or contextual factors compiled and reflected in a coherent way. For example, one participant explained that "seeing your statistics reflected back at you can be a really helpful tool... it's not making the decision for me," but helps assemble a broader picture of sleep, stress, or workload. Another noted that engaging in "a human-like discussion about it can sometimes put the whole thing together for you... I can better reflect on my state of mind at the moment." These uses were framed as self-directed and supplemental rather than diagnostic or directive.

When considering AI tools designed specifically for mental health support, more than half of participants (7/13) expressed conditional openness to AI-supported self-assessment. Some described potential value in systems that could help establish a baseline or integrate multiple contextual factors such as fatigue, workload, and cumulative stress. One participant suggested that AI could provide graded feedback, such as "green, yellow, red," to help clarify whether further rest or mitigation strategies were warranted. Importantly, these functions were consistently described as advisory rather than authoritative. As one participant distinguished, "It depends on whether you're talking about giving something control or something giving you advice... something giving me advice and help, I welcome." AI was therefore viewed as acceptable when it supported reflection and awareness, but not when it replaced professional judgment.

At the same time, concerns about privacy, surveillance, and documentation strongly shaped anticipated engagement. A majority of participants (8/13) expressed hesitation about using AI for mental health if it were integrated into employer or regulatory systems. One participant asked directly, “Are they going to tell the company?... I don’t want what I say to be used against me.” Others described preferring to manage concerns privately rather than introduce written records into institutional systems. Participants also questioned the feasibility of mandatory AI-based risk assessments, emphasizing that any such tool would need to remain voluntary and under pilot control. Across accounts, AI was not rejected outright. Instead, its acceptability depended on whether it functioned as a trust-bounded, private outlet for reflection, rather as a formally recorded or institutionally embedded assessment mechanism.

4 DISCUSSION

4.1 Mental Health Support as Exposure-Sensitive Negotiation

Across interviews, pilots’ accounts closely aligned with prior literature on aviation mental health [1, 9], reaffirming well-documented concerns around disclosure risk, information traceability [4], and the normalization of constrained coping practices within safety-critical work. However, the present findings clarify that barriers to engagement are not driven by stigma alone. Rather, pilots consistently framed mental health support as exposure-sensitive: engagement decisions were shaped by how information could become professionally visible, formally documented, and potentially consequential for licensure and employment.

Mental health disclosure was evaluated through the lens of institutional visibility. Participants described carefully managing what became written, recorded, or formally attributable, often delaying or avoiding diagnosis to preserve control over documentation pathways. Even intermediary resources such as union-sponsored programs were assessed according to how effectively they insulated pilots from formal escalation. These accounts suggest that mental health support in aviation is negotiated within layered systems of regulatory, organizational, and operational constraint, where professional identity and employability remain tightly coupled to certification status.

The most analytically generative contribution of this study lies in how pilots reasoned about emerging AI-mediated support within these exposure-sensitive conditions. Rather than rejecting AI outright, participants articulated a conditional openness, grounded in where and how such systems would operate.

4.2 Positioning AI Within Layered Systems of Exposure

The layered socio-technical framework (Figure 1) synthesizes this tension between openness and hesitation. Outermost layers, regulatory and organizational systems, carry higher levels of documentation visibility and professional consequences. Operational demands generate sustained cognitive and emotional strain within these structures. The innermost layer represents the individual pilot’s personal and professional identity, where engagement remains comparatively private and self-directed.

Within this structure, AI-mediated mental health support was perceived as viable when positioned at the boundary between operational and performance demands and the individual pilot’s personal and professional identity. Participants described AI as useful for reflection, baseline self-assessment, and sensemaking, particularly when it integrated contextual factors such as fatigue, workload, and stress without assuming decision-making authority. AI was framed as advisory rather than authoritative, and as supportive rather than diagnostic.

However, this conditional openness was consistently accompanied by reservations regarding institutional integration. Participants expressed reluctance toward AI tools embedded within employer or regulatory systems, particularly when engagement could generate persistent records or formal documentation. Under current trust conditions, the integration of AI into organizational or certification processes risks undermining engagement altogether. The framework therefore highlights not only whether AI is acceptable, but where it can operate without collapsing trust.

Rather than positioning AI as a solution to structural constraints, these findings suggest that its viability depends on alignment with existing exposure dynamics. AI-mediated support becomes meaningful when it functions as a trust-bounded, non-institutional outlet under pilot control, and becomes problematic when it extends into formally documented or surveillance-adjacent systems.

4.3 Limitations

This study is limited by the small number of participants included in the analysis ($n=13$). Although additional interviews were available from UMelb, these data were excluded due to differences in study focus and interview protocols. Participants were based across Asia, Europe, and North America, and findings primarily reflect pilots trained and operating within Western regulatory constraints, limiting generalizability to pilots in non-Western or underrepresented regions.

4.4 Future Direction

This study examined pilots’ perceptions of AI-mediated mental health support rather than direct interaction with such systems. Future research should also examine how the placement of AI within or outside institutional systems shapes trust and sustained engagement. Specifically, through in-situ, observational, or longitudinal studies, and explore how trust, privacy concerns, and perceived risk emerge through use over time. Comparative research across regulatory and organizational contexts, alongside more diverse pilot populations, may help identify conditions that better support trusted and effective AI-mediated mental health support.

5 CONCLUSION

Pilots’ engagement with mental health resources and AI-mediated support is shaped by concerns about information traceability, documentation, and institutional visibility. AI-mediated mental health tools were viewed as acceptable when they functioned as trust-bounded, self-directed resources positioned within the individual domain, rather than as extensions of organizational or regulatory oversight. These findings suggest that successful adoption of tailored AI-mediated mental health support in aviation depends less

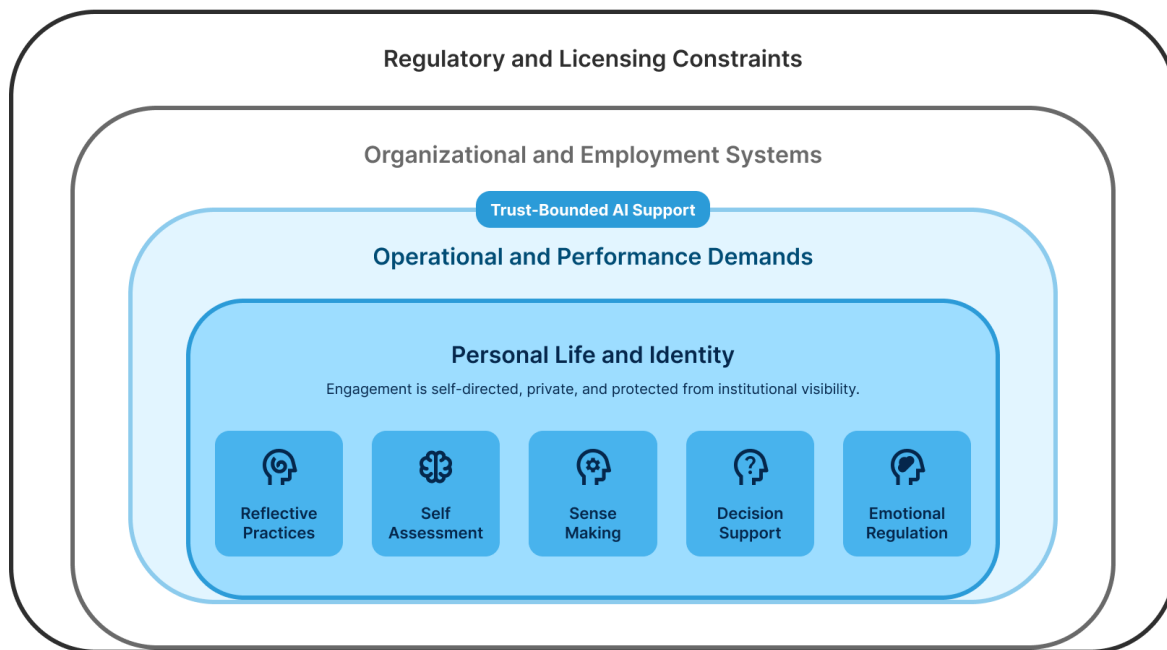


Figure 1: A layered socio-technical framework illustrating pilots' nested systems of institutional exposure. Outermost layers (regulatory and organizational) carry higher documentation visibility and professional consequence, while operational demands generate ongoing strain within these systems. The innermost layer, personal life and identity, represents a private, self-directed domain protected from institutional oversight. Trust-bounded AI support operates at the boundary between operational demands and personal life, functioning as a private outlet without extending into organizational or regulatory systems.

on attitudes toward technology and more on how such tools are positioned within documentation-sensitive professional systems. Designing AI that remains non-institutional and under pilot control may be essential for sustaining trust and engagement.

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References

- [1] Joan Cahill, Paul Cullen, Sohaib Anwer, Simon Wilson, and Keith Gaynor. 2021. Pilot Work Related Stress (WRS), Effects on Wellbeing and Mental Health, and Coping Methods. *The International Journal of Aerospace Psychology* 31, 2: 87–109. <https://doi.org/10.1080/24721840.2020.1858714>
- [2] Kathy Charmaz. 2012. *Constructing grounded theory: a practical guide through qualitative analysis*. Sage, Los Angeles.
- [3] Shane Cross, Imogen Bell, Jennifer Nicholas, Lee Valentine, Shaminka Mangelsdorf, Simon Baker, Nick Titov, and Mario Alvarez-Jimenez. 2024. Use of AI in Mental Health Care: Community and Mental Health Professionals Survey. *JMIR Mental Health* 11: e60589–e60589. <https://doi.org/10.2196/60589>
- [4] William R. Hoffman, James Aden, R. Daniel Barbera, Ryan Mayes, Adam Willis, Parth Patel, and Anthony Tvaranas. 2022. Healthcare Avoidance in Aircraft Pilots Due to Concern for Aeromedical Certificate Loss: A Survey of 3765 Pilots. *Journal of Occupational & Environmental Medicine* 64, 4: e245–e248. <https://doi.org/10.1097/JOM.00000000000002519>
- [5] Kevin W Jin, Qiwei Li, Yang Xie, and Guanghua Xiao. 2023. Artificial intelligence in mental healthcare: an overview and future perspectives. *The British Journal of Radiology* 96, 1150: 20230213. <https://doi.org/10.1259/bjr.20230213>
- [6] Han Li, Renwen Zhang, Yi-Chieh Lee, Robert E. Kraut, and David C. Mohr. 2023. Systematic review and meta-analysis of AI-based conversational agents for promoting mental health and well-being. *npj Digital Medicine* 6, 1: 236. <https://doi.org/10.1038/s41746-023-00979-5>
- [7] Kimberly Perkins, Rachael Merola, and Tasnim Hasan. 2025. From Awareness to Action: Mapping Emotional Intelligence to Pilot Performance and Policy Reform in Aviation Mental Health. <https://doi.org/10.54941/ahfe1006991>
- [8] Christine Schillings, Echo Meißner, Benjamin Erb, Eileen Bendig, Dana Schultchen, and Olga Pollatos. 2024. Effects of a Chatbot-Based Intervention on Stress and Health-Related Parameters in a Stressed Sample: Randomized Controlled Trial. *JMIR Mental Health* 11: e50454. <https://doi.org/10.2196/50454>
- [9] Marion Venus, Danja Greder, and Martin Grosse Holtforth. 2022. How professional pilots perceive interactions of working conditions, rosters, stress, sleep problems, fatigue and mental health. A qualitative content analysis. *European Review of Applied Psychology* 72, 3: 100762. <https://doi.org/10.1016/j.erap.2022.100762>
- [10] Alexander C. Wu, Deborah Donnelly-McLay, Marc G. Weisskopf, Eileen McNeely, Theresa S. Betancourt, and Joseph G. Allen. 2016. Airplane pilot mental health and suicidal thoughts: a cross-sectional descriptive study via anonymous web-based survey. *Environmental Health* 15, 1: 121. <https://doi.org/10.1186/s12940-016-0200-6>
- [11] Yihao Zheng, Dina Kaur Chawla, and Kimberly Perkins. 2025. Digital Copilots: Advancing Pilot Mental Health Through AI Chatbots and Systems. <https://doi.org/10.54941/ahfe1007021>